

RumourEval

COMP4167: Natural Language Processing

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1 Analytics & Exploratory Data Analysis

1.1 1(a) N-gram & Token Analysis

To analyse how stance is expressed linguistically, I computed the most frequent unigrams and bigrams in reply tweets for each stance label (Support, Deny, Query, Comment). After basic pre-processing (lower-casing and stopword removal), token frequencies were extracted using a bag-of-words representation. The resulting top unigrams and bigrams for each class are presented in Figures 1 and 2.

Top Unigrams in Replies by Label (Train)

Label S			Label D		
Rank	N-gram	Count	Rank	N-gram	Count
1	police	97	1	not	65
2	hostages	87	2	no	25
3	paris	76	3	flag	22
4	hostage	74	4	don't	19
5	sydney	74	5	just	14
6	breaking	55	6	like	14
7	cafe	54	7	what	13
8	#charliehebd	54	8	that's	12
9	#ferguson	50	9	news	12
10	#sydneyseige	42	10	isis	12
11	killed	42	11	police	11
12	french	41	12	#ferguson	11

Label Q			Label C		
Rank	N-gram	Count	Rank	N-gram	Count
1	what	34	1	not	208
2	why	30	2	no	138
3	not	20	3	what	118
4	no	19	4	just	118
5	police	18	5	police	108
6	how	16	6	people	97
7	did	13	7	don't	86
8	does	11	8	like	86
9	source	11	9	why	73
10	where	10	10	know	68
11	sydney	10	11	hostages	67
12	right	9	12	#ferguson	59

Figure 1: Top Unigrams in Replies by Label (Train).

Top Bigrams in Replies by Label (Train)

Label S			Label D		
Rank	N-gram	Count	Rank	N-gram	Count
1	sydney cafe	33	1	not isis	6
2	charlie hebdo	25	2	isis flag	5
3	supermarket paris	25	3	that's not	5
4	french alps	21	4	islamic flag	5
5	thomas samson	19	5	parliament hill	4
6	hostages freed	19	6	did not	4
7	samson #afp	19	7	don't think	4
8	freed jewish	19	8	not islamic	4
9	paris photo	19	9	not true	4
10	photo thomas	19	10	don't know	3
11	jewish supermarket	19	11	not playing	3
12	hostage situation	18	12	don't care	3

Label Q			Label C		
Rank	N-gram	Count	Rank	N-gram	Count
1	why did	5	1	parliament hill	25
2	sydney cafe	4	2	kosher market	14
3	why not	3	3	looks like	14
4	how know	3	4	surge pricing	13
5	why does	3	5	shots fired	13
6	looks like	3	6	hostage taking	13
7	isis flag	3	7	shooting hostage	12
8	need wait	2	8	market paris	12
9	mike brown	2	9	taking kosher	12
10	michael brown	2	10	don't know	12
11	how police	2	11	paris reported	11
12	hostages escape	2	12	stay safe	11

Figure 2: Top Bigrams in Replies by Label (Train).

N-gram:

- **Support (S)** replies often repeat factual details from the source (e.g. place names like “Sydney” or event terms such as “#CharlieHebdo”), indicating alignment with the rumour.
- **Deny (D)** replies are characterised by explicit negation and correction, with frequent cues such as “not” and phrases like “that’s not” or “not isis”, signalling rejection.
- **Query (Q)** replies are dominated by interrogative terms (e.g. “what”, “why”, “why did”), reflecting requests for clarification and uncertainty.
- **Comment (C)** replies contain more generic language (e.g. “just”, “like”) and event-related bigrams (e.g. “parliament hill”, “kosher market”), which reference the rumour without expressing a clear stance.

Overall, n-gram patterns clearly differentiate stance classes: support focuses on facts, denial on negation, queries on questions, and comments on general conversation.

Reply/Source Token Distribution:

Figure 3 compares token distributions between source and reply texts for stance (S/D/Q) and non-stance (C) classes.

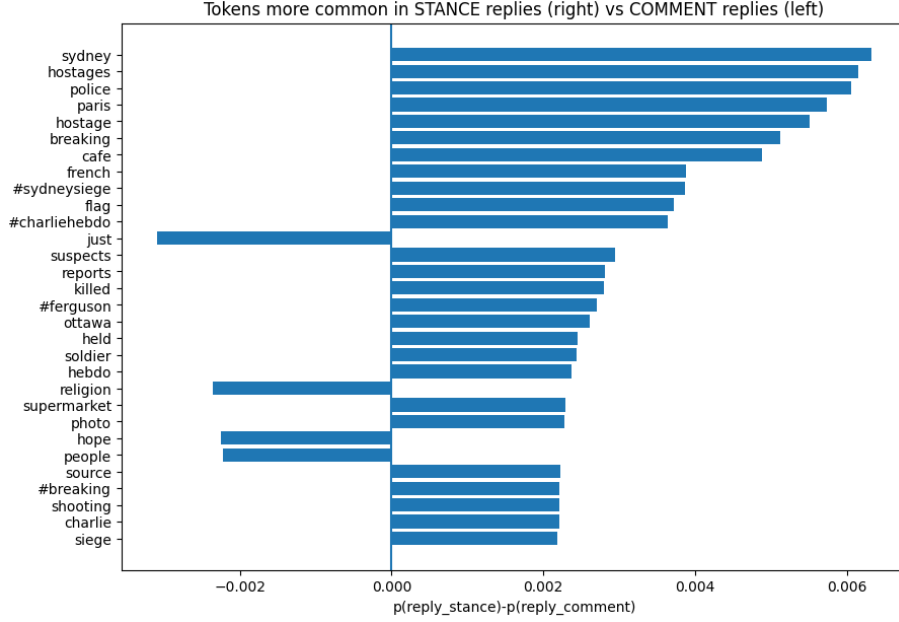


Figure 3: Contrastive difference of empirical token probability distributions.

Figure 3 compares token distributions between stance-bearing (S/D/Q) and non-stance (C) replies using conditional probability differences. Stance replies are dominated by event-specific and factual tokens (e.g. locations, entities, and breaking-news hashtags), indicating direct engagement with the rumour. In contrast, Comment replies contain more generic and conversational language (e.g. “just”, “people”, “hope”), reflecting social reaction rather than explicit stance. Overall, stance is expressed through distinct lexical cues rather than topic alone.

1.2 1(b) LDA Tobic Analysis

To analyse thematic differences between stance-bearing and non-stance replies, I applied Latent Dirichlet Allocation (LDA) separately to replies labelled as stance (Support, Deny, Query) and Comment. For each group, the eight topics were extracted and visualised using mean topic proportions and word clouds based on top topic words. This analysis focuses on comparing thematic focus between stance and non-stance replies, rather than distinguishing individual stance categories.

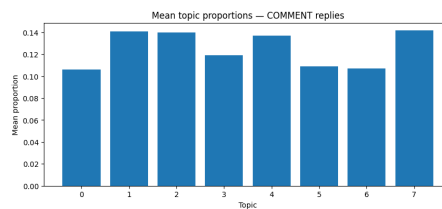
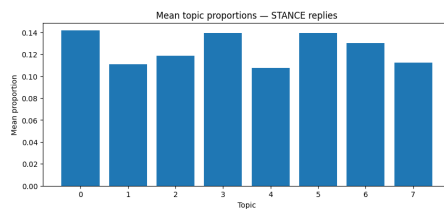


Figure 4: Visualisation of topic word lists: mean topic proportions (STANCE).

Figure 5: Visualisation of topic word lists: mean topic proportions (COMMENT).

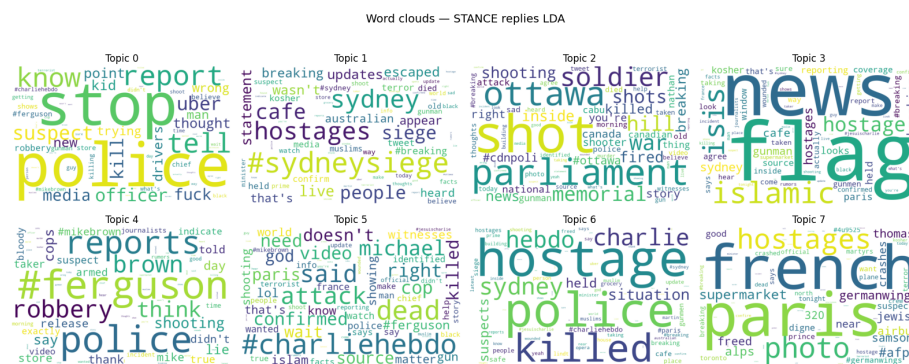


Figure 6: Word Cloud: STANCE replies LDA

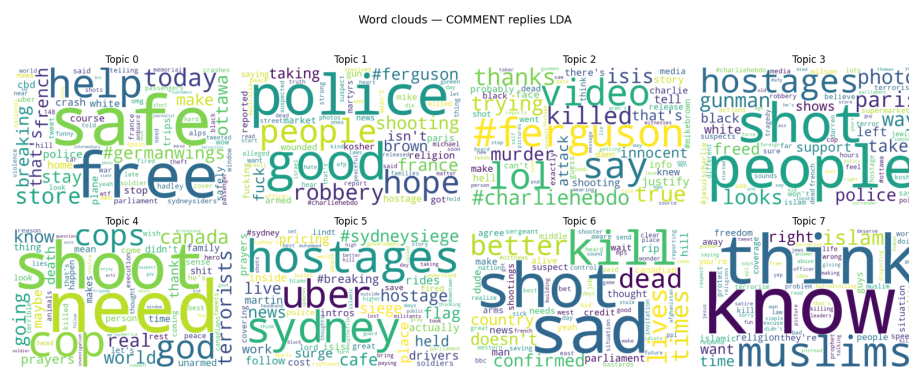


Figure 7: Word Cloud: COMMENT replies LDA

Figures 4–5 show that topic proportions are similarly distributed across stance and Comment replies, with only small differences, but with complementary emphasis across topics. In contrast, the word clouds (Figures 6–7)

reveal clearer linguistic differences: stance replies use more varied and personalised language, while Comment replies rely on more frequently repeated terms, reflecting rumour propagation rather than critical engagement.

1.3 Q1 Evaluation

While n-gram and LDA analyses highlight systematic lexical and thematic differences between stance-bearing and Comment replies, these methods rely on bag-of-words assumptions and ignore context and discourse structure. In particular, LDA yields coarse-grained topics that do not distinguish between Support, Deny, and Query replies, meaning the observed differences should be treated as suggestive rather than definitive.

2 2(a) 4-way Classification

2.0.1 2(a)(I) Model Training, Design Choices, and Diagnostics

Model and Input Formulation: For 4-way stance classification (S, D, Q, C), a transformer model pre-trained on Twitter data was fine-tuned. In particular, we selected **BERTweet** due to its pre-training on large-scale English tweets, making it well-suited to the RumourEval dataset. Each instance was represented as a **paired input** consisting of the *source-tweet* and its corresponding *reply-tweet*.

Preprocessing: For preprocessing, we took necessary steps to align with BERTweet’s pre-training conventions while preserving relevant linguistic stance cues. URLs were normalised to HTTPURL, user mentions were replaced with @USER, and excess whitespace was removed. No aggressive token filtering or stopword removal was applied, as negation markers, interrogatives, and pragmatic tokens (e.g. “why”, “how”, question marks) are highly informative for stance detection. This lightweight normalisation reduces lexical sparsity without discarding cues relevant to D and Q.

Imbalance Mitigation: The dataset is highly imbalanced (C=2291, S=743, D=273, Q=262). To mitigate this, class-weighted cross-entropy was used, with weights computed via square-root inverse frequency to reduce majority dominance without inducing instability. Macro-averaged F1 was chosen as the primary metric to equally weight minority stance performance.

Training Setup and Hyperparameters: We fine-tuned the model on the provided training split, with early stopping based on *dev-macro-F1* to reduce overfitting. We used *AdamW* with weight decay and a linear learning rate schedule with warm-up for optimisation. For regularisation, we used gradient clipping, increased dropout and label smoothing to reduce over-confidence. To

ensure reproducibility, we used fixed random seeds. Alongside the experiment outputs, we exported all hyperparameters, configurations and weights.

Training Diagnostics: Figure 8 shows the **training loss curve**, which shows a consistent decrease throughout fine-tuning, indicating stable optimisation and learning. It also shows that further improvement remained marginal after 1000 or so steps.

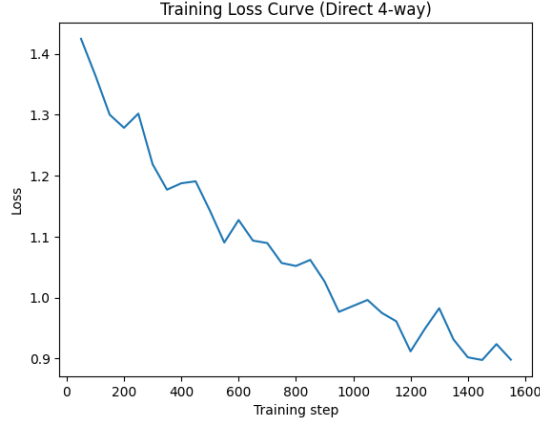


Figure 8: Training loss (direct 4-way) across training steps.

Early stopping selected the checkpoint with the highest development macro-F1 ([**DEV MACRO-F1 = 0.535, EPOCH=5.0, ACCURACY = 0.670**]), showing moderate, but meaningful performance across classes. The macro F1 shows that the model learns useful signals for minority stances, particularly for *Support* (F1 = 0.527) and *Query* (F1 = 0.591).

2.0.2 2(a)(II) Test-set Performance and Imbalance Analysis

Overall Performance: In contrast, the test performance highlights severe generalisation issues since it drops to a **macro-F1** of [**TEST MACRO-F1 = 0.228**] and an overall accuracy of [**TEST ACCURACY = 0.724**] with low recall. This shows that on the test set, the model has over-fit to the majority class, showing majority class-bias under the distribution shift. This discrepancy demonstrates that class imbalance and limited transferability of stance cues substantially hinder single-stage 4-way classification, motivating more structured modelling approaches. Per-class precision, recall, F1-score, and support values are reported in Figure 9.

DEV per-class metrics					TEST per-class metrics				
Class	Precision	Recall	F1	Support	Class	Precision	Recall	F1	Support
S	0.569	0.491	0.527	167	S	0.000	0.000	0.000	94
D	0.225	0.282	0.250	71	D	0.000	0.000	0.000	71
Q	0.588	0.594	0.591	96	Q	0.152	0.047	0.072	106
C	0.771	0.776	0.773	616	C	0.742	0.969	0.841	778

Figure 9: Direct 4-way Classification: Per-class Precision / Recall / F1 / Support

Effect of Class Imbalance: Class imbalance strongly affects test performance. The model overwhelmingly predicts *Comment*, resulting in near-zero recall for other classes. The skewed predicted label distribution shows that minority stance cues do not generalise reliably under distribution shift in our current model. The data shows that most stance-bearing replies are misclassified as *Comment*, with minimal confusion between stance categories, reflecting the implicit nature of stance cues in short replies.

Summary: Overall, the direct 4-way classifier performs reasonably on the development set but fails to generalise under severe class imbalance on the test set. Dominance of the *Comment* class leads to strong majority-class bias and poor minority-class recall despite explicit imbalance mitigation, indicating that a single-stage 4-way formulation is insufficient and motivating more structured modelling approaches in later sections.

3 2(b) Generative/ LLM-based Prompting

3.1 Model and Setup

I used **google/flan-t5-large** as the generative model. Each instance concatenates the **source** and **reply** tweets, and the model predicts a single label code. I use **ranked prompting**, selecting the argmax over the token log-probabilities of A,B,C,D, which enforces deterministic outputs and avoids free-form or invalid generations. To reduce truncation risk, tokenisation uses **left truncation**. Additionally, I clip long texts to fixed character budgets inside the prompt, unfortunately this will reduce the Macro-F1.

Few-shot examples were sampled from the training set, selecting replies with clear stance cues (enquiry or question marks for *Query*, explicit negation for *Deny*, and agreement terms for *Support*). On the DEV split, zero-shot prompting achieved a **macro-F1 of 0.4278**. Few-shot prompting performed best with $k = 1$ example per class (macro-F1 = 0.4012), while larger k values degraded

performance sharply (macro-F1 ≈ 0.07), indicating that additional demonstrations can confuse our simple model.

3.2 Evaluation

Development set: I evaluate on the DEV set to choose the recommended prompt design, then report TEST performance. The main metrics in consideration are the macro-averaged F1’s. Surprisingly, zero-shot outperformed few-shot on DEV and is therefore the recommended prompt design.

DEV set.

	DEV macro-F1
Zero-shot (ranked)	0.4552
Few-shot (ranked, k=1)	0.4138

When looking at per-class behaviour, we can see the main weakness: **Deny is extremely low-recall**. For zero-shot, Support and Query are moderately captured (F1 ≈ 0.53 and 0.46), Comment is strongest (F1 ≈ 0.75), but Deny collapses (F1 ≈ 0.09). Few-shot did worse: Deny degraded further (F1 ≈ 0.04), and the model predicted Comment even more often (658 predictions vs. 616).

Test set. Both prompts failed to generalise:

	TEST macro-F1
Zero-shot (ranked)	0.2129
Few-shot (ranked, k=1)	0.2129

A macro-F1 of 0.2129 is consistent with a degenerate strategy that predicts only the majority class (*Comment*).

Transferable Bias: A lightweight calibration step was applied on DEV by adding bias offsets to the log-scores and tuning them to maximise DEV macro-F1. Calibration slightly improved few-shot DEV performance (**macro-F1 = 0.4167**) but had no effect on TEST (macro-F1 = 0.2129 for both prompting modes). This indicates that the performance gap is not due to a simple transferable bias.

Qualitative Analysis The dominant failure mode is a complete collapse to the majority class:

- **Default-to-comment rate (TEST, zero-shot):** 1.0
- **Minority→Comment errors (TEST, zero-shot):** 271

This occurred despite explicit instructions not to default to the majority class, indicating that prompt constraints alone are insufficient under distribution shift. Even on DEV, *Deny* recall is near zero. Many *Deny* replies are subtle (e.g.

sarcasm or indirect contradiction) and do not align with the explicit keyword-style cues encoded in the prompt, making this class particularly sensitive to imbalance and implicit language. Increasing few-shot k sharply degraded performance, despite remaining within the token limit, suggesting that additional demonstrations introduce conflicting patterns or encourage over-generalisation.

Recommendation: Based on DEV macro-F1, the recommendations include:

- use **Zero-shot** over Few-shot.
- **Ranked prompting** rather than free-form generation.

Despite reasonable (but low) DEV performance (macro-F1 ≈ 0.46), both prompt variants collapsed to the majority class on TEST, indicating that prompt-only classification with a general instruction-tuned model is not robust under severe imbalance and distribution shift. This motivates the use of supervised fine-tuning.

4 2(c) Chain-of-Thought

4.1 Two-Stage Pipeline

Stage 1: Comment vs Non-Comment: We fine-tune a BERTweet as a binary classifier on the full training set, relabelling S/D/Q as **Non-Comment**. We achieved a **DEV Macro-F1 = 0.662**.

At inference, Stage 1 outputs $p(\bar{C} \mid x)$ and $p(C \mid x)$. A threshold τ is applied:

$$\hat{y}_1(x) = \begin{cases} \text{Comment} & \text{if } p(C \mid x) \geq \tau \\ \text{Stance} & \text{otherwise} \end{cases}$$

Stage 2: Support vs Deny vs Query: Stage 2 is a separate BERTweet classifier trained only on S/D/Q). This removes the majority class entirely and reallocates capacity to the subtle distinctions among minority stances.

Stage 2 DEV macro-F1 = 0.636

End-to-end 4-way prediction: Final prediction is obtained by chaining the stages:

$$\hat{y}(x) = \begin{cases} \text{Comment} & \text{if } \hat{y}_1(x) = \text{Comment} \\ \hat{y}_2(x) \in \{\text{Support}, \text{Deny}, \text{Query}\} & \text{otherwise} \end{cases}$$

where $\hat{y}_2(x) = \arg \max_{k \in \{S, D, Q\}} p(k \mid x, \text{NonC})$ is the Stage 2 argmax.

Method	DEV macro-F1	TEST macro-F1
Direct 4-way BERTweet (2a)	0.54	0.23
Prompting (zero-shot) (2b)	0.46	0.21
Two-stage (Two-Stage)	0.52	0.21

Table 1: Overall 4-way macro-F1 comparison.

4.2 Results and Comparison to Baselines

Table 1 shows that the two-stage classifier improves DEV macro-F1 by separating **Comment** detection from **S/D/Q** classification, allowing Stage 2 to learn stronger minority-class distinctions. On the test set, the two-stage model is more robust than both baselines but is still constrained by Stage 1 routing errors. This indicates that chain-of-thought decomposition improves robustness under imbalance, but does not fully resolve generalisation issues.

4.3 Summary (and Tradeoffs)

Table 1 shows that, although the two-stage chain-of-thought model is theoretically better suited to severe class imbalance, it does not outperform the direct 4-way classifier in practice. While the two-stage approach benefits from explicitly separating **Comment** detection from **S/D/Q** classification, its end-to-end performance is bottlenecked by Stage 1, limiting overall macro-F1. In contrast, the direct 4-way model, despite being conceptually simpler, avoids this hard routing constraint and therefore performs slightly better overall.

These results should be interpreted in the context of the dataset and task difficulty. The training data is extremely small for a four-way semantic classification problem (Train: 3569, Dev: 950, Test: 1049 instances) and highly imbalanced ([TRAIN] C=2291, S=743, D=273, Q=262), making robust generalisation challenging. Under such data scarcity and restricted computational budgets, the additional complexity of the two-stage pipeline amplifies error propagation rather than consistently improving performance.

Overall, the results show that chain-of-thought methods require reliable intermediate predictions to be effective; under severe data scarcity and imbalance, simpler end-to-end models can outperform more complex pipelines by avoiding error propagation.

5 3 - Ethical Considerations

Data Collection, Consent, and Privacy: Although tweets are publicly available, users did not explicitly consent to their content being used for research beyond the original platform context. Reliance on Twitter’s terms of service may not align with user expectations, particularly as tweets can express personal opinions or emotional reactions. Replies may also contain sensitive information such as names, locations, or speculative accusations, raising privacy concerns, especially for generative approaches. In this project, tweets are treated purely as

text, with no user-level metadata or identity inference, and outputs are restricted to stance labels. Any real-world deployment would require additional safeguards such as anonymisation and strict data minimisation.

Bias: The dominance of the *Comment* class biases models towards majority predictions, under-representing denial and questioning. This is mitigated by prioritising macro-averaged F1 and per-class metrics, alongside class weighting and a two-stage classification strategy to improve minority-class recall.

Model Behaviour and Misinterpretation Risks: Stance classification may be misinterpreted as fact verification, as labels reflect user reactions rather than truth. If treated as credibility signals, predictions could amplify misinformation in high-stakes domains. Accordingly, stance detection is framed as conversational analysis with closed-set outputs, though real-world use would still require fact-checking and human oversight.

Potential for Future Misuse: At scale, stance detection could be misused for surveillance or viewpoint suppression, particularly if systematic misclassification affects minority groups. Responsible use therefore requires transparency about limitations and separation from enforcement or policy decisions.